

**EXPERIMENT – 13 - EMBEDDED C PROGRAM TO ADD AN ARRAY
OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN
INTERNAL RAM**

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PROGRAM

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {
```

```
        0x1111,
```

```
        0x2222,
```

```
        0x8888,
```

```
        0x4444,
```

```
        0xABCD
```

```
    };
```

```
    unsigned long sum = 0;
```

```
    for(i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    while(1);
```

OUTPUT

